Market-Implied FOMC Probabilities from Fed Funds Futures

An end-to-end methodology guide

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1 The goal and the big idea

Goal. For every trading day t in the sample, and for every FOMC meeting scheduled after t, compute the market-implied probability of each possible policy outcome (hold, ±25bp, ±50bp, ...) and the resulting distribution of the target range.

Big idea in three sentences.

1. A fed funds futures contract settles on the *average* overnight rate over one calendar month, so its price reveals the market's expected average rate for that month.
2. The overnight rate only changes at FOMC meetings, so each monthly average is a known, day-weighted blend of the rates before and after the meetings in that month. Un-blending the averages gives the expected rate after each meeting.
3. An expected rate is a probability-weighted average of possible outcomes. If outcomes are restricted to a 25bp grid and to the two grid points nearest the expectation, the expectation pins down a unique probability.

Intuition

Think of each contract as a betting slip on "the average rate in month j". You observe 13 such slips. You know the rate can only jump on meeting days. From the slips you reverse-engineer how big a jump the market expects on each meeting day, and then you translate "expected jump" into "probability of jump".

2 Inputs and notation

2.1 Data

Series	Bloomberg	Role in the method
Fed funds futures, months 1–13	`FF1–FF13 Comdty`	Expected monthly average rates
Effective fed funds rate	`FEDL01 Index`	Starting rate; realized days of the current month
Target range upper bound	`FDTR Index`	Grid of outcomes; validation
Target range lower bound	`FDTRFTRL Index`	Grid of outcomes; EFFR spread
FOMC decision dates	Fed website	Where the rate path can jump

2.2 Notation

Symbol	Meaning
t	Observation date (one row of the data)
$j = 0, 1, \ldots, 12$	Calendar month relative to t ($j = 0$ is the month containing t)
N_j	Number of calendar days in month j
$P_{t,n}$	Price of contract FFn on date t; FFn refers to month $j = n - 1$
$\bar{r}_{t,j}$	Implied average rate for month j, in percent
$r_t(s)$	Overnight rate on calendar day s (realized or expected)
$k = 1, \ldots, K_t$	FOMC meetings with decision date after t, in order
$D_k,\ j_k,\ d_k$	Decision date of meeting k, its month index, its day of the month
$r_k^-,\ r_k^+$	Expected rate before and after meeting k
Δ_k	Expected change at meeting k
h	Step size of policy moves, $h = 0.25$ percentage points
x_k	Expected change in units of steps, $x_k = \Delta_k / h$
$\pi_k(i)$	Probability that, after meeting k, the target is i steps from today's range

All rates are in percent. All day counts are *calendar* days, because the overnight rate applies on weekends and holidays at the previous business day's level.

3 Step 0 — Clean the data

1. **Restrict the sample** to the period of interest (here 2021 onward). Earlier data has stale prints in the far contracts.
2. **Drop exchange-holiday rows.** With weekday sampling and fill-previous, holidays appear as copies of the previous day. When a holiday falls on the first day of a month, the row carries the *previous* month's contracts under a new-month date, which breaks Step 1.
3. **Convert to numbers** and treat `#N/A` as missing.

Why this step

Every later step assumes that on date t, contract FF1 refers to the month containing t. A single mislabeled row produces a nonsense probability on that day.

4 Step 1 — Map each contract to its calendar month

Bloomberg's generic contracts roll on the first business day of each month, so on date t:

$$\text{FF}n \;\longleftrightarrow\; \text{month } j = n-1 \text{ relative to } t, \qquad n = 1, \ldots, 13. \tag{1}$$

For example, on 15 September 2026, FF1 is September 2026, FF2 is October 2026, FF4 is December 2026.

Why this step

The un-blending in Step 4 needs to know which meetings fall inside which contract's month. That requires an exact contract-to-month mapping for every date.

5 Step 2 — From price to expected monthly average

The contract is quoted as 100 minus the rate, so

$$\bar{r}_{t,j} \;=\; 100 - P_{t,\,j+1}. \tag{2}$$

At expiry the contract settles on the arithmetic average of the daily effective rate over the month. Before expiry, its price is (to a close approximation) the market's expectation of that average:

$$\bar{r}_{t,j} \;\approx\; \mathbb{E}_t^{\mathbb{Q}}\left[\frac{1}{N_j}\sum_{s\in\text{month } j} r(s)\right]. \tag{3}$$

The superscript $\mathbb{Q}$ is a reminder that this is a *market-implied* (risk-neutral) expectation, which can include a small risk premium.

6 Step 3 — Model the rate path as a step function

6.1 The assumption

Between meetings the overnight rate is assumed flat. It can only jump at a meeting, and the new rate takes effect the day *after* the decision:

$$r(s) = \begin{cases} r_k^- & s \le D_k, \\ r_k^+ & D_k < s \le D_{k+1}. \end{cases} \tag{4}$$

Consecutive meetings are chained: the rate after one meeting is the rate before the next,

$$r_{k+1}^- = r_k^+, \qquad r_1^- = r_0, \tag{5}$$

where r_0 is the rate in force today (Step 6 discusses how to measure it).

6.2 What each monthly average looks like

Substituting the step function into (3) gives three cases.

(a) A month with no meeting. The rate is constant all month, equal to the rate after the most recent meeting:

$$\bar{r}_{t,j} = r_k^+ \quad \text{where } k \text{ is the last meeting before month } j. \tag{6}$$

(b) A future month with one meeting on day d. The old rate applies for d days and the new rate for $N - d$ days:

$$\bar{r}_{t,j} \;=\; \frac{d}{N}\, r_k^- \;+\; \frac{N-d}{N}\, r_k^+. \tag{7}$$

(c) The current month ($j = 0$), partly realized. Days up to τ (the last day with a published effective rate) are already known. Only the remaining days are expectations:

$$N_0\, \bar{r}_{t,0} \;=\; \underbrace{\sum_{s=1}^{\tau} r^{\text{obs}}(s)}_{R_\tau \text{ (realized)}} \;+ (d-\tau)\, r_1^- \;+\; (N_0 - d)\, r_1^+. \tag{8}$$

If the realized days were all at r_1^-, this collapses back to (7).

Intuition

Each monthly contract is a weighted average of at most two unknowns (before and after the meeting), with known weights given by the calendar. That turns a pricing problem into simple algebra.

7 Step 4 — Solve for the rate after each meeting

Work forward through the meetings $k = 1, 2, \ldots, K_t$. For each meeting there are two ways to get r_k^+.

Method A: un-blend the meeting month. Rearranging (7):

$$r_k^+ \;=\; \frac{N\, \bar{r}_{t,j_k} \;-\; d_k\, r_k^-}{N - d_k}. \tag{9}$$

For the current month use (8) instead: $r_1^+ = \big(N_0 \bar{r}_{t,0} - R_\tau - (d-\tau)\, r_1^-\big)/(N_0 - d)$.

Method B: anchor on the next month. If the month after the meeting has no meeting, (6) gives the answer directly:

$$r_k^+ \;=\; \bar{r}_{t,\, j_k+1}. \tag{10}$$

7.1 Which method to use

Method A amplifies price noise. Differentiating (9):

$$\frac{\partial r_k^+}{\partial \bar{r}_{t,j_k}} \;=\; \frac{N}{N - d_k}. \tag{11}$$

For a meeting on the 16th of a 30-day month the factor is about 2. For a meeting on the 28th of a 31-day month it is about 10: a 1bp pricing error becomes a 10bp error in the post-meeting rate, which is a 40-percentage-point error in probability.

Rule. Use Method B whenever the month after the meeting contains no meeting. Otherwise use Method A. In both cases set $r_{k+1}^{-} = r_k^{+}$ and move to the next meeting.

Why this step

This is the core "un-blending" step. It converts 13 monthly averages into a path of expected post-meeting rates, one per meeting, using only the calendar and the prices.

Advanced alternative. All months can be written at once as a linear system $\bar{r} = A\,\mathbf{r}^{+} + b$, where row j of A holds the day-weights of each post-meeting rate in month j. With 13 months and up to about 10 meetings the system is over-determined and can be solved by least squares. This uses all contracts but needs care with weighting.

8 Step 5 — Expected move per meeting

$$\Delta_k = r_k^{+} - r_k^{-} \qquad \text{(incremental: this meeting only)} \tag{12}$$

$$C_k = r_k^{+} - r_0 = \sum_{m=1}^{k} \Delta_m \qquad \text{(cumulative: from today)} \tag{13}$$

Express both in steps of $h = 0.25$:

$$x_k = \frac{\Delta_k}{h}, \qquad c_k = \frac{C_k}{h} = \sum_{m=1}^{k} x_m. \tag{14}$$

For example, $x_k = 0.92$ means "0.92 of a 25bp hike is priced at meeting k".

9 Step 6 — The starting rate and the EFFR spread

The futures settle on the *effective* rate, but outcomes are defined on the *target range*. Define the spread

$$s_t \;=\; \mathrm{EFFR}_t - \frac{U_t + L_t}{2}, \tag{15}$$

where U_t, L_t are the target bounds. Two practical points:

- **Incremental moves do not depend on the spread** if it is constant over the horizon: it shifts every r by the same amount and cancels in Δ_k. So Steps 7–8 are spread-free.
- **The starting rate r_0 does depend on it.** Use a robust recent EFFR (for example, the median over the last 5–10 business days, excluding decision days), not a single day's print.

When measuring s_t, skip decision days. On those days the bounds update immediately but the effective rate only moves the next day, creating a false spike. The spread is not constant across regimes (about -4.5bp in 2021–2024, about $+1$bp in 2026), so use a rolling estimate.

10 Step 7 — From expected move to probabilities (one meeting)

10.1 The assumption

Policy moves come in multiples of h. At each meeting only the two multiples bracketing the expectation are possible:

$$n_k = \lfloor x_k \rfloor, \qquad \text{outcomes} \in \{\, n_k h,\ (n_k+1)h \,\}. \tag{16}$$

10.2 The derivation

Let p_k be the probability of the larger move. Matching the expectation:

$$h\big[(1-p_k)\,n_k + p_k\,(n_k+1)\big] = \Delta_k \quad \Longrightarrow \quad h\,(n_k+p_k) = h\,x_k. \tag{17}$$

$$p_k = x_k - \lfloor x_k \rfloor, \qquad \Pr\big(\text{move} = (n_k+1)h\big) = p_k, \qquad \Pr\big(\text{move} = n_k h\big) = 1-p_k. \tag{18}$$

Examples.

- $x = 0.92$: $n = 0$, $p = 0.92$. That is 92% hike 25bp, 8% hold.
- $x = -0.72$: $n = -1$, $p = 0.28$. That is 72% cut 25bp, 28% hold.
- $x = 2.6$: $n = 2$, $p = 0.6$. That is 60% hike 75bp, 40% hike 50bp.

Intuition

The futures give one number per meeting, so they can identify only one probability. Restricting to the two nearest outcomes is the narrowest distribution consistent with the price. It is a convention, not a measurement of tail risk.

11 Step 8 — The full distribution across meetings (probability tree)

11.1 Recursion

Let i count 25bp steps away from today's target range. Start with all mass on today's range and apply each meeting's two-point move to every branch:

$$\pi_0(i) = \mathbf{1}\{i = 0\}, \tag{19}$$

$$\pi_k(i) = (1-p_k)\,\pi_{k-1}(i-n_k) \;+\; p_k\,\pi_{k-1}(i-n_k-1). \tag{20}$$

State i after meeting k corresponds to the target range

$$\big[\,L_t + i\,h,\ \ U_t + i\,h\,\big]. \tag{21}$$

11.2 Consistency property

Each step adds $n_k + p_k = x_k$ to the mean, so

$$\mathbb{E}_{\pi_k}[\,i\,] = \sum_{m=1}^{k} x_m = c_k. \tag{22}$$

The distribution therefore always reproduces the futures-implied cumulative move. This is a useful automatic check.

11.3 Zero lower bound

If $L_t + i\,h < 0$, merge that mass into the $[0, 0.25]$ range. Report this explicitly, because it breaks the consistency property above.

11.4 Alternative: two-point on levels (WIRP style)

Instead of the tree, apply Step 7 directly to the cumulative move c_k:

$$\Pr\big(i = \lfloor c_k \rfloor + 1\big) = c_k - \lfloor c_k \rfloor, \qquad \Pr\big(i = \lfloor c_k \rfloor\big) = 1 - \big(c_k - \lfloor c_k \rfloor\big). \tag{23}$$

This puts mass on only two levels at every horizon. It is simpler and matches what WIRP displays, but it understates dispersion at longer horizons. The tree matches the CME FedWatch convention. Choose one and use it consistently.

Why this step

Per-meeting probabilities answer "what happens at this meeting?". The tree answers "where will the rate be after this meeting?", which is what you need for questions about rate *levels* at a date.

11.5 Caveat

The recursion (20) applies the same move probabilities to every branch. In reality the chance of a hike in October probably depends on whether September hiked. Futures carry no information about that dependence, so the tree is a convention.

12 Step 9 — Build the time series

Repeat Steps 1–8 for every date. A meeting is *upcoming* on date t only if $D_k > t$. Futures settle after the 2pm ET announcement, so on a decision day the decision is already in the price and that meeting must be excluded. For unscheduled meetings, include them only from the day they were announced.

Output (long format). One row per (date, meeting, outcome):

Algorithm 1 Implied FOMC probabilities for one date t

$\bar{r}_{t,j} \leftarrow 100 - P_{t,j+1}$ for $j = 0, \ldots, 12$ ▷ Step 2
Meetings $\leftarrow \{k : D_k > t,\ j_k \le 12\}$ ▷ Step 1 and 9
$r^- \leftarrow r_0$ (robust recent EFFR) ▷ Step 6
$\pi \leftarrow \{0 : 1\}$
for each meeting k in date order **do**
    if month $j_k + 1$ has no meeting and $j_k + 1 \le 12$ **then**
        $r^+ \leftarrow \bar{r}_{t,j_k+1}$ ▷ Method B
    else
        $r^+ \leftarrow \big(N\,\bar{r}_{t,j_k} - d_k\,r^-\big)/(N - d_k)$ ▷ Method A (or eq. 8 if $j_k = 0$)
    end if
    $x_k \leftarrow (r^+ - r^-)/h;\ \ n_k \leftarrow \lfloor x_k \rfloor;\ \ p_k \leftarrow x_k - n_k$ ▷ Steps 5, 7
    $\pi \leftarrow$ apply (20) with (n_k, p_k) ▷ Step 8
    record $(t, D_k, r^+, \Delta_k, C_k, p_k, \pi)$
    $r^- \leftarrow r^+$
end for

Date t	Meeting D_k	Type	Outcome	Probability
2026-09-15	2026-09-16	move at meeting	+25bp	0.921
2026-09-15	2026-09-16	move at meeting	hold	0.079
2026-09-15	2026-10-28	level after meeting	3.75–4.00	0.518
...	...	...	...	...

From this table you can build fixed-meeting series (how the September probability evolved day by day), rolling series (probability of a move at the next meeting), and event studies.

13 Worked example: 15 September 2026

The day before the September 2026 meeting. Target range $[3.50, 3.75]$, EFFR = 3.63 (and every September day so far printed 3.63).

Inputs

Contract	Month	Price	$\bar{r}$ (%)	Meeting in month
FF1	Sep 2026	96.2625	3.7375	16 Sep ($d = 16$, $N = 30$)
FF2	Oct 2026	96.1250	3.8750	28 Oct ($d = 28$, $N = 31$)
FF3	Nov 2026	96.0200	3.9800	none
FF4	Dec 2026	95.8950	4.1050	9 Dec ($d = 9$, $N = 31$)

Meeting 1 — 16 September

The month after (October) has a meeting, so use Method A. All realized days were at 3.63, so (8) reduces to (9):

$$r_1^+ = \frac{30 \times 3.7375 - 16 \times 3.63}{14} = 3.8604, \qquad \Delta_1 = 23.0\text{bp}, \quad x_1 = 0.921.$$

So $n_1 = 0$ and $p_1 = 0.921$: **92.1% hike 25bp, 7.9% hold.** (The Fed hiked 25bp the next day.)

Meeting 2 — 28 October

November has no meeting, so use Method B:

$$r_2^+ = \bar{r}_{\text{Nov}} = 3.98, \qquad \Delta_2 = 3.98 - 3.8604 = 12.0\text{bp}, \quad x_2 = 0.479.$$

So **47.9% hike 25bp, 52.1% hold.**

For comparison, Method A on the October contract gives $r_2^+ = (31 \times 3.875 - 28 \times 3.8604)/3 = 4.012$. That is 3bp away from the anchor, because the amplification factor is $31/3 \approx 10$. This is exactly why the rule prefers Method B.

Meeting 3 — 9 December

January has a meeting, so use Method A on December:

$$r_3^+ = \frac{31 \times 4.105 - 9 \times 3.98}{22} = 4.1561, \qquad \Delta_3 = 17.6\text{bp}, \quad x_3 = 0.705.$$

So **70.5% hike 25bp, 29.5% hold.**

Distribution of the target range (tree)

After meeting	3.50–3.75	3.75–4.00	4.00–4.25	4.25–4.50	Mean hikes
16 Sep	7.9%	92.1%	—	—	0.92
28 Oct	4.1%	51.8%	44.1%	—	1.40
9 Dec	1.2%	18.2%	49.5%	31.1%	2.10

Check: the mean number of hikes in each row equals the cumulative $c_k = 0.92,\ 1.40,\ 2.10$, as the consistency property requires.

14 Validation checklist

1. **Probabilities are valid:** $0 \le p_k \le 1$ and $\sum_i \pi_k(i) = 1$ for every date and meeting.
2. **Mean matches futures:** $\sum_i i\,\pi_k(i) = c_k$ (except where the zero lower bound binds).
3. **Day-before test:** on the day before each meeting, the implied probability of the actual decision should usually be high.

4. **Method agreement:** where both Method A and Method B are available, they should agree within a few bp. Large gaps flag bad prices.
5. **External benchmark:** on a handful of dates, compare with Bloomberg WIRP or CME FedWatch history. Differences of a few percentage points are normal.

15 Assumptions and limitations

1. **Risk-neutral, not real-world.** Prices include a risk premium, small for near meetings and growing with horizon.
2. **25bp grid, two outcomes per meeting.** Tail outcomes (for example 50bp emergency moves) are not identified. Options would be needed for that.
3. **Constant spread** between EFFR and the range midpoint over the horizon.
4. **Same move probabilities on every branch** of the tree.
5. **No intermeeting moves** anticipated in prices.
6. **Liquidity** declines with maturity, so the far contracts are noisier.

16 Comparing with prediction-market prices

If the goal is to compare these probabilities with event contracts on FOMC outcomes:

- **Match outcome definitions.** Map the buckets above to the contract's resolution rule (for example, change in the upper bound, or the level of the range after the meeting).
- **Match timestamps.** Futures prices here are daily settlements. Use the prediction-market price at the same time of day.
- **Remember the measure difference.** Futures give risk-neutral probabilities. Prediction-market prices also embed fees, spreads and their own participants' risk preferences.

A Formula summary

Implied average:	$\bar{r}_{t,j} = 100 - P_{t,j+1}$
Meeting month:	$\bar{r}_{t,j} = \frac{d}{N} r_k^- + \frac{N-d}{N} r_k^+$
Solve (Method A):	$r_k^+ = \dfrac{N\bar{r}_{t,j_k} - d_k r_k^-}{N - d_k}$
Anchor (Method B):	$r_k^+ = \bar{r}_{t,j_k+1}$ if month $j_k + 1$ has no meeting
Chain:	$r_{k+1}^- = r_k^+, \quad r_1^- = r_0$
Expected move:	$\Delta_k = r_k^+ - r_k^-, \quad x_k = \Delta_k / 0.25$
Two-point:	$n_k = \lfloor x_k \rfloor, \quad p_k = x_k - n_k$
Tree:	$\pi_k(i) = (1 - p_k)\pi_{k-1}(i - n_k) + p_k\, \pi_{k-1}(i - n_k - 1)$
Check:	$\sum_i i\, \pi_k(i) = \sum_{m \le k} x_m$